

AI Vocal Coach – Ilysia Krzywonos, Chris Zhao

Project Overview

The primary goal of this project is a vocal coach that will provide real-time feedback on singing techniques by analyzing pitch accuracy, loudness control, vibrato quality, and breathiness. The goal is to create an accessible tool that helps singers improve through data-driven feedback.

Data Collection

- **GTSinger** will be the primary source for pitch tracking as in NanoPitch, as well as annotations for vibrato, note transition frames, and breath marks. We may also extract chest/head/mixed/falsetto voice labels.
- **VocalSet** will be referenced for additional vibrato & phonation variation data, and as the primary source for dynamic level annotations.
- Smule's **DAMP** dataset may also be referenced for a larger corpus of amateur-level singing.

Feature Extraction

- Extract **fundamental frequency** (f_0) with model built off of current NanoPitch and analytically score pitch drift over the course of a held note
- **Gesture classification**: An additional model head will classify each frame as steady, vibrato, glissando, or transition based on GTSinger's annotations.
 - Steady frames will use f_0 to determine pitch accuracy or drift.
 - Vibrato and glissando frames will be scored using analytic measures.
 - Transition frames unscored to allow for transitional pitch dips.
- **Vibrato analysis**: Compute a short mean and use signal processing techniques to determine vibrato frequency, amplitude, and mean. This mean will be used to score pitch accuracy. Vibrato consistency will be evaluated throughout the song.
- **Breath control**: A light CNN will be trained to recognise breath events from the spectrogram based on GTSinger's annotations, which will break up singing into phrases. Cepstral peak prominence & spectral tilt will be computed analytically to evaluate breathiness consistency through phrases & the song.

- **Loudness and phrasing:** Loudness will be computed analytically through RMS and classified as *piano*, *mezzopiano*, *mezzoforte*, *forte*, etc. through VocalSet's annotations, and phrase boundaries will be marked through breath events. Within each phrase, a regression head will evaluate the shape of the envelope to score loudness consistency.
- **Register detection:** Chest / mixed / head / falsetto voice classification head + feedback for unintentional breaks may be added as an additional measure.

Model

- **GRU Stack:** We will build off an improved version of NanoPitch's architecture to determine f_0 as well as gesture classification (vibrato/transition/etc.), dynamics & phrasing, and potentially register classification. The Viterbi decoding will then be modified to analytically account for vibrato and glissando shape before scoring.
- **Breath CNN:** We will train a small convolution neural network to recognise breath events directly from the raw waveform for phrase segmentation.

User Interface

- Raw output:
 - **f_0 pitch** and **central pitch** plotted on the same time graph
 - **Vibrato** activation, onset, frequency, amplitude, visualized as a wave
 - **Breath event marking** & abstract **breathiness level** through CPP
 - **Dynamic classification & loudness graph** by breaths (phrase visualization)
 - **Register** (chest / mixed / head / falsetto)
- Feedback:
 - **Pitch accuracy** (based on equal temperament) and **drift** (across sustained notes)
 - **Vibrato rate/depth** within reasonable range and **stability** across notes & song
 - **Breathiness consistency** based on CPP & spectral tilt across notes/phrases
- UX features:
 - Pitch, vibrato, breathiness, dynamics, and register are hideable/expandable cards
 - Post-processed summary to avoid overwhelming real-time feedback
 - Compute numerical scores based on professional vs amateur examples (?)



AI VOCAL COACH ARCHITECTURE

From Data to Insights to Actionable Feedback

